

Observer-Relative Closure Signatures on Replay Artifact Graphs: A Bounded Source-Facing Audit of Existing LLM Artifacts

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May 6, 2026

Abstract

We present Gate12B, a read-only secondary audit for studying observer-relative closure signatures in already materialized Gate12A inference transport artifacts. A closure signature is treated here as a relation-patterned candidate surface, not as a correctness label or model-quality score. Across four dense-transformer model lines, archive-family Gate12B surfaces repeatedly place high-tension candidates on `residual_chord=3` and flat candidates on `residual_chord=1|trusted_tree=2`. The archive-family direction is observed under the representative motif path and the scale-focused support path. Under the core-observer boundary path, candidates can vanish, marking a boundary of the default observer set. The artifact-level reparameterization record is reported as a bounded availability-and-outcome record, not as independent support. Under the selected source-facing queue reported in this paper, high-side R candidates align with `conflict-following` rows in 8/8 cases, and flat-side M candidates align with `support-following` or `non-gluing` rows in 8/8 cases. Transcript and briefing sensitivities do not reproduce the same clean archive alignment. We present the result as a bounded artifact report, not as a universal interpretability law, correctness classifier, model-quality benchmark, weight-level causal claim, or physical invariant.

1 Introduction

LLM evaluation often compresses generated output into scalar scores or task-level labels. Those views are useful, but they discard much of the structure present in intermediate audit artifacts. This report takes a different stance. It studies existing replay artifacts as structured records: local objects, directed transport relations, explicit triangle cycles, and source-facing text surfaces are inspected without rerunning model inference and without converting the result into a model ranking.

The artifact line used here is Gate12A/Gate12B. Gate12A emits discrete inference transport artifacts. Gate12B reads those artifacts as a secondary audit and asks whether relation-patterned closure signatures remain visible when the same candidate surface is viewed through different observer views, coarse-grained scale modes, and a bounded local reparameterization check. Gate12B does not alter Gate12A semantics, thresholds, classifications, or source text. It is an artifact reader.

The Gate12B surface is downstream of the controlled structural replay regime reported in [1] and the transport-first closure formulation in [2].

Informally, Gate12B asks whether the same triangle-level non-closure patterns keep reappearing after changing which view of the artifact is used, how the cycles are grouped, and how the local representation is rewritten within the bounded transform available in the artifact package. The report therefore studies persistence across artifact views, not free-form model behavior. The object of interest is a candidate structural surface between raw generated text and aggregate benchmark scores.

The main observation is an archive-family result. In the current dense-transformer artifact set, archive-family Gate12B surfaces repeatedly place high-tension candidates on `residual_chord=3` and flat candidates on `residual_chord=1|trusted_tree=2`. The same queue and source-facing annotation path does not make transcript and briefing surfaces show the same clean archive alignment. This contrast matters because it reduces the risk that the archive result is merely a consequence of the annotation rule or the motif observer expansion.

This report contributes:

1. a read-only secondary audit framing for existing inference transport artifacts;
2. an observer and scale sensitivity surface for closure signatures, together with a bounded artifact-level reparameterization availability-and-outcome record;
3. a bounded archive-family relation-signature result across four dense-transformer model lines;
4. source-facing queue and derived annotation evidence connecting selected structural candidates back to text surfaces;
5. non-archive sensitivities showing that transcript and briefing do not reproduce the same clean archive alignment.

The paper follows a deliberately conservative path. Section 2 positions the report relative to adjacent interpretability and evaluation work. Section 3 defines the artifact regime. Section 4 gives operational definitions for external readers. Section 5 describes the Gate12B read-only audit and candidate construction. Section 6 reports the archive-family relation-signature flip, sensitivity checks, source-facing annotation, and non-archive sensitivity. Section 7 explains why the result matters as an artifact-level audit signal. Section 8 states threats to validity. Section 9 maps claims to artifacts and integrity checks. The conclusion restates the bounded claim and the non-claims.

1.1 What We Claim

1. **Bounded artifact phenomenon.** We report a repeated archive-family closure signature in existing Gate12A/Gate12B audit artifacts.
2. **Archive-family relation signature.** Across four dense-transformer model lines, archive-family Gate12B surfaces repeatedly place high-tension candidates on `residual_chord=3` and flat candidates on `residual_chord=1|trusted_tree=2`.

3. **Sensitivity boundary.** The archive-family direction is observed under the representative motif path and the scale-focused support path. The core-observer boundary path is reported as a setting where candidates can vanish, and the artifact-level reparameterization record is reported as a bounded availability-and-outcome record, not as independent support.
4. **Source-facing alignment.** A selected archive source-facing queue aligns high-side R candidates with **conflict-following** rows and flat-side M candidates with **support-following** or **non-gluing** rows.
5. **Non-archive sensitivity.** Transcript and briefing sensitivities do not reproduce the same clean archive alignment under the same queue and derived annotation path.

1.2 What We Do NOT Claim

1. **No universal law.** We do not claim a universal law of LLM reasoning.
2. **No benchmark ranking.** We do not present Gate12B as a model-quality benchmark or correctness classifier.
3. **No answer-quality labels.** Source-facing tags describe the direction of a queued source row only; they are not answer-quality labels.
4. **No weight-level claim.** We make no claim about model weights in general or about causal mechanisms inside a model.
5. **No physical invariant.** The bounded reparameterization check is an artifact-level stability check over the evaluated transform, not a physical invariant.
6. **No Gate12A checkpoint claim.** This report is not a Gate12A checkpoint or replacement for Gate12A.

2 Positioning

This report is adjacent to mechanistic interpretability, geometric deep learning, and holistic model evaluation [3, 4, 5, 6, 7], but it does not attempt to localize circuits, probe hidden activations, or rank models by task performance. It studies already materialized audit artifacts: replay graphs, transport relations, triangle cycles, residual bands, and source-facing text joins. The closest conceptual neighbors are structural audit, replay analysis, and artifact-level evaluation rather than claims about model-internal mechanisms.

The practical gap is between raw text inspection and aggregate scoring. Raw generated text is inspectable but hard to compare reproducibly across artifact views. Scalar scores are comparable but can erase the local transport and closure structure recorded by the audit pipeline. Gate12B is positioned as a middle audit layer: it preserves relation-patterned candidate surfaces long enough for observer, scale, and source-facing sensitivity checks before any human reader decides what the row means.

3 Controlled Artifact Regime

The empirical surface is deliberately narrow. Gate12B reads existing Gate12A artifacts and emits secondary audit summaries. No model inference is run for Gate12B. No Gate12A artifact schema, threshold, classification, or source semantics is changed.

The selected model lines are:

- Qwen/Qwen2.5-0.5B
- Qwen/Qwen2.5-3B-Instruct
- meta-llama/Llama-3.2-3B-Instruct
- Qwen/Qwen3-4B

The selected rendering families are:

- archive_v1
- transcript_v1
- briefing_v1

Archive is the family in which the repeated closure signature is reported. Transcript and briefing are sensitivity surfaces used to test whether the same structural and source-facing pattern is reproduced outside archive.

Table 1: Artifact regime and fixed study surface.

Dimension	Values and boundary
Model lines	Qwen/Qwen2.5-0.5B; Qwen/Qwen2.5-3B-Instruct; meta-llama/Llama-3.2-3B-Instruct; Qwen/Qwen3-4B
Families	archive_v1; transcript_v1; briefing_v1
Source artifacts	Gate12A discrete connection artifacts and Gate12A triangle text-surface audit artifacts where source-facing rows are used
Secondary audit	Gate12B observer-relative coarse-grained closure audit; read-only over Gate12A outputs
Representative motif path	observer_mode_set=cycle_motif_expansion_v1; top_k=3; min_observer_support=3; min_scale_support=3
Source queue setting	per_band_limit=2; selected queue rows only; derived exact-anchor / phrase-rule annotation

4 Definitions and Terminology

This section fixes the vocabulary used in the report. The definitions are operational. They describe how the artifact readers work; they are not philosophical or semantic claims about reasoning in

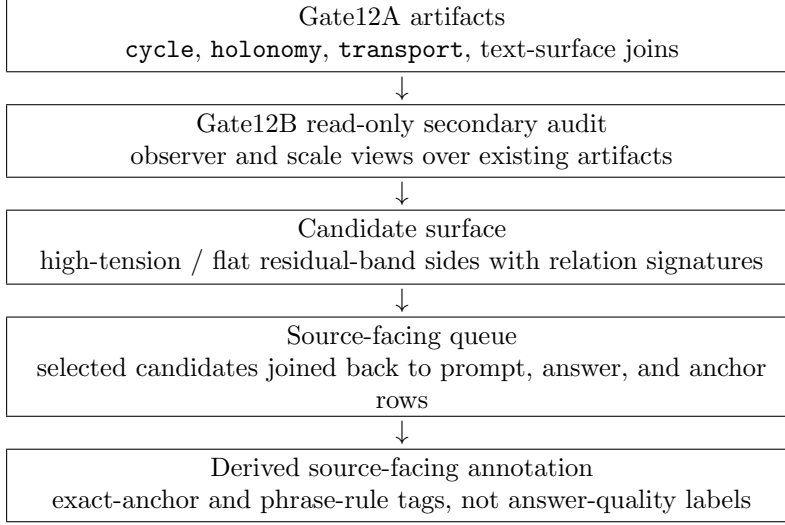


Figure 1: Read-only Gate12B audit flow. The pipeline starts from existing Gate12A artifacts and returns selected structural candidates to text surfaces without running new model inference or creating a model-quality score.

general.

Gate12A artifact. A Gate12A artifact is a materialized audit bundle containing local proxy objects, directed transport relations, explicit triangle cycle registries, closure residuals, and manifests. For source-facing inspection, a matching Gate12A text-surface audit supplies prompt, answer, support-anchor, and conflict-anchor fields joined to triangle-cycle rows.

Gate12B secondary audit. Gate12B is a read-only artifact reader over Gate12A outputs. It builds observer-relative and coarse-grained views of closure residuals, emits candidate rows that satisfy support requirements, and checks a bounded local reparameterization condition when array-level transport operators are present.

Local object. A local object is the artifact-level representation used by Gate12A to attach replay state to nodes and transport relations. This report does not interpret local objects as model-internal variables.

Transport relation. A transport relation is a directed artifact edge connecting local objects in the replay graph. In the selected Gate12A surface, relations can be summarized by relation kinds such as `trusted_tree` and `residual_chord`.

Trusted-tree edge. A `trusted_tree` relation is an edge belonging to the trusted-tree part of the selected Gate12A relation decomposition. It is used as a relation kind in triangle signatures; it is not a correctness label.

Residual-chord edge. A `residual_chord` relation is a residual edge in the selected Gate12A relation decomposition. It is a relation kind used in cycle signatures; it is not an answer-quality label by itself.

Triangle cycle. A triangle cycle is an explicit three-leg cycle in the Gate12A cycle registry. Gate12B uses these cycles as the base surface for closure residuals and relation-signature grouping.

Closure signature. In this report, a closure signature is not a scalar score and not a correctness judgment. It is a relation-patterned candidate surface observed after applying Gate12B support requirements across observer and scale views.

Relation signature. A relation signature summarizes the relation kinds appearing on a selected triangle path. The shorthand `residual_chord=3` means that all three legs in the selected triangle signature are residual-chord relations. The shorthand `residual_chord=1|trusted_tree=2` means that one leg is a residual-chord relation and two legs are trusted-tree relations. These strings should not be overinterpreted as semantic labels.

Candidate. A Gate12B candidate is a cycle row that survives candidate construction requirements. It is not simply a top-k triangle. A candidate must have support across exact-deduplicated observer views, support across multiple scale modes, and at least one non-triangle coarse-scale support mode.

High-tension and flat. High-tension and flat are residual-band sides used by the candidate extractor. They are candidate sides, not correctness classes. A high-tension row may be source-facing conflict-following, but that does not make high-tension a general failure label. A flat row may be source-facing support-following or non-gluing, but that does not make flat a general correctness label.

Observer. An observer is an artifact-level view used to condition which cycles participate in the candidate surface. Gate12B groups observer views with identical cycle membership before counting support. Observer support is an implementation-level count over distinct observer views after exact-set deduplication, not a statistical independence claim. Observer views are merged only when their selected `cycle_id` sets are exactly equal; highly overlapping but non-identical views remain distinct in the current report.

Scale and coarse-graining. A scale mode groups the same triangle rows at different levels of aggregation. The current runner includes triangle-level scale plus relation-kind, anchor-policy, and residual-quantile scale modes. Candidate rows must be supported across enough scale modes and must include at least one non-triangle coarse-scale mode.

Bounded local reparameterization. The current reparameterization check is an artifact-level stability check over the evaluated transform. In the current implementation, the array-level transform is `basis_coordinate_reversal_v1`; a registry-only identity fallback is not promoted as evidence for nontrivial reparameterization stability. This is a bounded check, not a claim of full coordinate invariance.

Source-facing queue. A source-facing queue joins selected Gate12B candidate rows back to prompt, answer, support-anchor, and conflict-anchor text-surface rows. The queue is an inspection surface, not a new model run.

Source-facing annotation. Source-facing annotation is derived from the queue by exact-anchor and phrase rules. It assigns `support-following`, `conflict-following`, `non-gluing`, or `ambiguous`. It is not human semantic annotation, not an answer-quality label, and not a model benchmark.

5 Method

5.1 Inputs

Gate12B reads existing Gate12A registries:

- `manifest.json`
- `explicit_triangle_cycle_registry.jsonl`
- `triangle_holonomy_registry.jsonl`
- `transport_relation_registry.jsonl`

When source-facing evidence is needed, the source inspection queue also reads matching Gate12A triangle text-surface artifacts:

- `triangle_text_surface_joined.jsonl`
- text-surface `manifest.json`

The queue builder requires the Gate12B run and the text-surface audit to share the same `source_gate12a_run_id`. This prevents a candidate surface from being silently joined to text rows from a different Gate12A source artifact.

5.2 Observer Axis

The default observer mode set is `core_v1`. It contains the core Gate12B observer views used by the initial runner:

- `all_edges`
- `anchor_qualified`
- `residual_chord_heavy`
- `relation_kind_conditioned`

The motif sensitivity uses `cycle_motif_expansion_v1`. This mode set keeps the core views and adds ordered cycle-motif views:

- `residual_first_leg`

- `residual_second_leg`
- `residual_third_leg`

The expansion is explicit. It is not a silent replacement for `core_v1`. Its role is to test whether the archive relation-signature result remains visible under the representative motif path once ordered cycle-motif views are counted as distinct observer views after exact-set deduplication where their cycle memberships differ.

Observer support is counted after deduplicating observer views by cycle membership. Two observer views are treated as the same exact-deduplicated observer view when the set of selected `cycle_id` values is identical for the candidate surface. This is exact set equality, not a similarity threshold.

5.3 Scale Axis

The current scale modes are:

- `triangle`
- `relation_kind_band`
- `anchor_policy_band`
- `residual_quantile_band`

Scale support is counted across the scale modes in which a cycle appears in the top-k flat or high-tension surface for an exact-deduplicated observer view. Candidate rows must meet `min_scale_support`; they also must have at least one non-triangle coarse-scale support mode. This prevents a row from being promoted only because it appears at triangle resolution.

5.4 Residual-Band Construction

Gate12B uses residual bands to define the two candidate sides used throughout this report. The high-tension side is selected from the high-residual side of the supported artifact surface, and the flat side is selected from the low-residual side. These sides are computed after grouping rows by exact-deduplicated observer view and scale mode; they are not semantic classes and not answer-quality labels.

The paper therefore uses *high-tension* and *flat* only as extractor-side residual-band names. A high-tension candidate is not by itself a failed answer, and a flat candidate is not by itself a correct answer. The source-facing annotation layer later asks how selected candidates read when returned to prompt, answer, support-anchor, and conflict-anchor surfaces.

5.5 Candidate Construction

Candidate construction has three key constraints. First, the row must be selected on a candidate side: flat or high-tension. Second, it must meet exact-deduplicated observer-view support. Third, it must meet scale support and include at least one coarse-scale mode. Observer views with identical cycle membership are reported together but counted as one exact-deduplicated observer view.

The representative motif path used for the paper tables is:

```

observer_mode_set    cycle_motif_expansion_v1
top_k                3
min_observer_support 3
min_scale_support    3
per_band_limit       2 for source queues

```

Table 2: Named support paths used in the report.

Path	Observer mode set	top_k	min_observer_support	min_scale_support
Representative motif path	cycle_motif_expansion_v1	3	3	3
Scale-focused support path	cycle_motif_expansion_v1	1/3/5	2	3
Core-observer boundary path	core_v1	1/3/5	3	2/3

These candidates are candidate signals only. They do not classify answers as right or wrong, and they do not score models.

5.6 Candidate Extraction Summary

The candidate extraction path is:

1. read the Gate12A triangle, holonomy, and transport registries;
2. construct observer views from the selected observer mode set;
3. merge observer views with identical selected `cycle_id` sets into one exact-deduplicated observer view;
4. evaluate flat and high-tension top-k membership within each exact-deduplicated observer view and scale mode;
5. keep rows meeting the observer and scale support thresholds, including at least one non-triangle coarse-scale support mode;
6. split retained rows into the high-tension and flat residual-band sides used in the reported tables;

7. optionally join selected rows to source-facing prompt, answer, support-anchor, and conflict-anchor text surfaces.

5.7 Bounded Local Reparameterization Check

Gate12B records an artifact-level reparameterization check when array-level transport operators are available. The checked transform is **basis_coordinate_reversal_v1**. The check is stated in terms of projector-level and transport-level compatibility under that evaluated transform. It is intentionally narrower than any full coordinate-system claim.

If the array-level transport operator artifact is absent, the runner records a registry-only identity path rather than promoting a nontrivial stability claim. The paper therefore treats reparameterization sensitivity as bounded to the evaluated artifact transform.

Table 3: Reparameterization-check availability and outcome boundary.

Evidence group / family	nontrivial basis_coordinate_reversal_v1 checks	registry-only fallback cases	failed check rows	Narrative treatment
Motif runs, archive_v1	12/12 runs; 67,584 check rows	0/12 runs	0	zero failed rows; not independent support
Motif runs, transcript_v1	12/12 runs; 67,584 check rows	0/12 runs	0	non-archive sensitivity boundary
Motif runs, briefing_v1	12/12 runs; 105,600 check rows	0/12 runs	48	threshold-boundary rows; not candidate-level instability
Archive boundary runs	24/24 runs; 92,160 check rows	0/24 runs	0	boundary evidence for archive sensitivity paths

Table 3 reports availability and runner-recorded outcomes for the Gate12B run groups used by the paper evidence map. The archive and transcript groups record zero failed check rows. The briefing group records 48 failed check rows, which are treated here as threshold-boundary rows rather than candidate-level instability for the archive claim. These failed rows occur in the briefing sensitivity surface and are not used as archive-family support. In this table, a failed check row means that the runner recorded **stable=false** under **basis_coordinate_reversal_v1**; this combines residual-band changes and residual-delta threshold failures under the runner’s recorded criterion. The reparameterization record is therefore reported as a bounded availability-and-outcome record, not as independent support for the main archive relation-signature result. The row counts in Table 3 are derived from the Gate12B reparameterization summary artifacts in the run directories listed by the canonical manifest. The larger briefing denominator reflects the larger number of reparameterization check rows materialized in the recorded Gate12B **gauge_stability_summary.json** artifacts, not a different reparameterization criterion. If a registry-only identity fallback occurs in a future run, that fallback is not counted as nontrivial reparameterization evidence.

5.8 Source-Facing Queue and Annotation

The source queue returns selected Gate12B candidates to Gate12A text surfaces. Each queued row carries candidate metadata, relation paths, prompt text, answer text, support-anchor text,

and conflict-anchor text. The queue builder uses a bounded `per_band_limit=2` for the selected archive, transcript, and briefing source queues.

The derived annotation procedure is deterministic. A case means one selected source-queue build for a fixed model line, rendering family, and `source_gate12a_run_id`.

1. The source queue is built case by case. Within each case, candidates are grouped by candidate side and relation signature. The `per_band_limit=2` setting selects at most two rows per side/signature group after deterministic sorting by candidate side, relation signature, residual-side order, and `cycle_id`.
2. The queue materializes `answer_contains_support_anchor_text` and `answer_contains_conflict_anchor_text` using the answer, support-anchor, and conflict-anchor fields already present in the queue row. Matching is case-insensitive, collapses whitespace, and requires a non-empty normalized anchor string to occur in the normalized answer string. No semantic paraphrase matching is used.
3. If the conflict-anchor match flag is true, the row is tagged `conflict-following`.
4. If no conflict-anchor match is present and the support-anchor match flag is true, the row is tagged `support-following`.
5. If neither anchor match is present but the answer contains one of the recorded no-gluing / not-warranted phrase patterns, the row is tagged `non-gluing`.
6. Otherwise the row is tagged `ambiguous`.

The current annotation helper uses the following literal lower-cased phrase patterns for the `non-gluing` rule:

<code>no direct</code>	<code>not warranted</code>	<code>cannot be glued</code>
<code>cannot be inferred</code>	<code>cannot infer</code>	<code>separate ledgers</code>
<code>cannot support</code>	<code>does not warrant</code>	<code>does not establish</code>

The tags are mutually exclusive under this precedence order.

This rule is useful for source-facing grounding because it records whether the answer surface follows the support anchor, follows the conflict anchor, or declines a direct source merge. It is not human semantic annotation, not an answer-quality label, not a model benchmark, and not a general evaluator.

6 Results

6.1 Archive Relation-Signature Flip

Question. Does the archive family show a repeated relation-signature direction across the current dense-transformer artifact set?

Observation. Yes. Archive-family Gate12B surfaces repeatedly place the high-tension side on R and the flat side on M.

Boundary. This is an archive-family artifact result over the current model set. It is not a universal statement about archive prompts or LLM behavior.

The main archive-family relation-signature result is:

```
archive high-tension side  residual_chord=3
archive flat side          residual_chord=1|trusted_tree=2
```

This pattern is observed across the four dense-transformer model lines listed in Section 3.

Table 4: Archive surface-level relation summary under the representative motif path.

Family / surface	High-side relation	Flat-side relation	Scope and interpretation
archive_v1	residual_chord=3	residual_chord=1 trusted_tree=2	current archive model-line surface across four dense-transformer model lines; clean archive direction

Table 5: Selected source-queue relation summary under the representative motif path.

Family	Selected queue rows	High-side relation among selected rows	Flat-side relation among selected rows	Reading
transcript_v1	16	residual_chord=1 trusted_tree=2 in 6/8 selected high rows	residual_chord=3 in 6/8 selected flat rows	mostly reverse / mixed
briefing_v1	16	residual_chord=1 trusted_tree=2 in 6/8 selected high rows	residual_chord=3 in 6/8 selected flat rows	mostly reverse / mixed

These tables use different evidentiary units: Table 4 is a surface-level archive summary, whereas Table 5 summarizes selected source-queue rows. They are orientation tables, not a like-for-like family taxonomy.

The shorthand “R” in the rest of the paper refers to `residual_chord=3`. The shorthand “M” refers to `residual_chord=1|trusted_tree=2`. These are relation signatures, not semantic labels.

6.2 Scale-Focused Support Path

Question. Does the archive relation-signature direction remain visible when scale support is strengthened while observer support is set to a weaker threshold than the representative motif path?

Observation. Yes. Under the scale-focused support path, the archive relation direction remains high R and flat M at `top_k=1`, `top_k=3`, and `top_k=5`.

Boundary. The scale-focused support path preserves the relation-signature direction, not every stronger dominance statement. The result should be read as a relation flip, not as a universal flat-dominance claim.

The scale-focused support path uses `observer_mode_set=cycle_motif_expansion_v1`, `top_k=1/3/5`, `min_observer_support=2`, and `min_scale_support=3`. It preserves the relation-signature direction at `top_k=1`, `top_k=3`, and `top_k=5`. The scale-focused support path weakens flat-candidate dominance relative to the baseline archive run, but it does not erase the relation-side alignment: high-tension remains R and flat remains M across the current dense-transformer archive surfaces.

This check should not be read as a scale-only ablation against the representative motif path, because the observer-support threshold differs from that path.

This is why the result is described as a relation-signature flip rather than as a flat-dominance claim. The stronger statement is not earned by all support-path settings.

6.3 Core-Observer Boundary Path

Question. Does the archive signal remain supported under the default core observer set when observer support is raised?

Observation. Not by default. Under the core-observer boundary path, the default core observer set can make candidates vanish. The representative motif path restores support by adding ordered cycle-motif views.

Boundary. The motif expansion is a recorded sensitivity path, not a replacement for the default observer set.

The core-observer boundary path uses the default core observer set, the available archive boundary runs with `top_k=1/3/5`, `min_observer_support=3`, and `min_scale_support=2/3`. This path exposes an important boundary: raising observer support under the default core observer set can make archive candidates vanish. This is not treated as a failure of the archive relation pattern. It says that the default core observer set is too narrow to support the reported archive candidate definition.

The representative motif path uses `observer_mode_set=cycle_motif_expansion_v1`, `top_k=3`,

`min_observer_support=3`, and `min_scale_support=3`. It restores support by adding ordered cycle-motif views. Under that path, archive keeps high R and flat M. The expansion remains a sensitivity path, not a new default.

6.4 Motif Observer Specificity

Question. Does motif observer expansion merely force every family into the archive direction?

Observation. No. Archive preserves the high-R / flat-M direction, whereas transcript and briefing mostly show the reverse selected direction.

Boundary. This supports a bounded family-specific reading. It does not make transcript or briefing failed families.

The motif observer expansion does not force every family into the archive direction. In the representative motif-path summaries used for the paper tables, archive preserves the high-R / flat-M direction, while transcript and briefing mostly show the reverse relation-signature direction in the selected source-facing queues. The broader 36 motif Gate12B runs are recorded in the manifest as the supporting motif evidence surface. This matters because it makes the archive result harder to explain as a trivial artifact of adding motif observers.

6.5 Sensitivity and Boundary Summary

Table 6: Sensitivity and boundary summary.

Check	Purpose	Observed result	Boundary
Scale-focused support path	Test whether archive direction remains visible with <code>min_observer_support=2</code> , <code>min_scale_support=3</code> , and <code>top_k=1/3/5</code>	Archive high remains R and flat remains M	Not a scale-only ablation against representative motif path
Core-observer boundary path	Test exact-deduplicated observer-view support under the default core observer set	Candidates can vanish	Boundary of current core views, not a contradiction of the archive relation pattern
Representative motif path	Add ordered cycle-motif observer views explicitly with <code>min_observer_support=3</code> , <code>min_scale_support=3</code> , and <code>top_k=3</code>	Archive high R / flat M is observed; transcript/briefing are not forced into archive direction	Sensitivity path, not default replacement
Bounded reparameterization	Check evaluated local basis transform	Denominator and failed rows reported in Table 3	Availability-and-outcome record only; fallback cases are not nontrivial evidence
Source-facing queue	Return candidates to prompt/answer/anchor text surfaces	Archive selected rows align high R with conflict and flat M with support or non-gluing	Selected <code>per_band_limit=2</code> queue only
Non-archive annotation	Test annotation-rule-only explanation	Transcript/briefing do not reproduce clean archive alignment	Sensitivity surfaces, not failed families

The Qwen3-4B briefing threshold-boundary case (`triangle:000358`, `tense -> flat`, residual delta approximately $1.6653345369377348e-16$) is recorded as threshold-boundary behavior, not candidate-level instability, and is not used as archive-family support.

6.6 Source-Facing Archive Alignment

Question. Do selected archive structural candidates align with source-facing text surfaces?

Observation. Yes in the selected queue. High-side R rows are **conflict-following** in 8/8 rows, while flat-side M rows are **support-following** in 6/8 rows and **non-gluing** in 2/8 rows.

Boundary. The annotation is exact-anchor / phrase-rule derived and source-facing only. It is not a semantic evaluator or answer-quality label.

The selected archive source-facing queue contains 16 rows:

- 8 high-tension rows

- 8 flat rows
- per_band_limit=2

The derived archive annotation result is:

high-side residual_chord=3 conflict-following in 8/8 rows
flat-side residual_chord=1|trusted_tree=2 support-following in 6/8 rows
flat-side residual_chord=1|trusted_tree=2 non-gluing in 2/8 rows

The two flat non-gluing rows are useful rather than anomalous. They record a case where the answer surface does not follow the conflict anchor but also does not glue the available notes into a direct support conclusion. The tag therefore remains source-facing and neutral: it is not a failure label.

Table 7: Abbreviated source-facing examples from the selected archive queue.

Candidate side / signature	Support anchor	Conflict anchor	Answer surface and derived tag
high_tension / residual_chord=3	a directed path exists from Aster to Noble	a directed path exists from Noble to Aster	Given the disputed memo, a directed path exists from Noble to Aster; conflict-following
flat / residual_chord=1 trusted_tree=2	Ivory is an ancestor of Vivid	<i>none recorded</i>	Given the retrieved notes, Ivory is an ancestor of Vivid; support-following

Table 7 is abbreviated from selected archive queue rows recorded by the manifest, specifically queue ranks 11 and 1. It is included only to make the queue surface concrete; the paper-level counts remain those in Table 8.

Table 8: Selected source-facing derived annotation summary.

Family	Rows	Support	Conflict	Non-gluing	Am-big.	High-side relation	Flat-side relation	Interpretation
archive_v1	16	6	8	2	0	residual_chord=3	residual_chord=1 trusted_tree=2	clean high-R conflict / flat-M support-or-non-gluing
transcript_v1	16	10	6	0	0	residual_chord=1 trusted_tree=2 in 6/8	residual_chord=3 in 6/8	split high/flat in aggregate; relation direction mostly reverse
briefing_v1	16	12	4	0	0	residual_chord=1 trusted_tree=2 in 6/8	residual_chord=3 in 6/8	model-conditioned / mixed; not clean archive alignment

These annotation tags describe source-facing direction only. They are not semantic labels and not answer-quality labels.

6.7 Non-Archive Source-Facing Sensitivity

Question. Does the same queue and annotation path make transcript and briefing reproduce the clean archive alignment?

Observation. No. Transcript and briefing do not reproduce the archive high-R conflict / flat-M support-or-non-gluing alignment. Their selected relation direction is mostly reversed, with high-side M and flat-side R in 6/8 selected rows.

Boundary. This constrains the archive interpretation. It does not show that transcript or briefing are incorrect, noisy, or less valuable.

The non-archive sensitivity constrains only a narrow degenerate explanation: that the same derived annotation procedure mechanically forces every selected family into the archive high-R conflict / flat-M support-or-non-gluing pattern. If that were the case, transcript and briefing would also show the same clean alignment. They do not. This check does not isolate annotation effects from rendering-family differences, source-text differences, anchor availability, candidate composition, or selected queue rows.

Transcript has 16 annotated rows: 10 **support-following**, 6 **conflict-following**, 0 **non-gluing**, and 0 **ambiguous**. Its selected high side is dominated by M in 6/8 rows, while its selected flat side is dominated by R in 6/8 rows.

Briefing also has 16 annotated rows: 12 **support-following**, 4 **conflict-following**, 0 **non-gluing**, and 0 **ambiguous**. Its selected high side is dominated by M in 6/8 rows, while its selected flat side is dominated by R in 6/8 rows.

This supports a bounded family-specific reading. It does not imply that transcript or briefing are incorrect, noisy, or scientifically unimportant. They are sensitivity surfaces that constrain the interpretation of the archive result.

7 Why This Matters

The result suggests that some audit artifacts may contain reusable structural signals that are neither scalar benchmark scores nor free-form human judgments. Gate12B does not decide whether an answer is correct. Instead, it asks whether relation-patterned non-closure surfaces remain visible after changing recorded observer views, scale grouping, and the bounded artifact-level representation available in the current package.

This creates a middle layer for audit tooling. Raw generated text can be too unstructured to compare reproducibly, while aggregate scores can erase the transport and closure structure present in replay artifacts. Gate12B keeps the candidate surface structural, then returns selected rows to source-facing text for inspection. The archive-family result is therefore useful not because it solves interpretability in general, but because it demonstrates a bounded way to connect persistent closure signatures back to source rows without turning them into model-quality labels.

For practitioners, the value of this surface is not that it replaces existing evaluation metrics. Its value is that it preserves a structured audit trace that can be inspected before collapsing an artifact into a scalar score. A model answer, a replay graph, and a source-facing text row can disagree in ways that are hard to express as accuracy, pass/fail, or preference labels. Gate12B keeps those disagreements visible as relation-patterned candidate surfaces.

This makes the result useful as an audit primitive. It can identify candidate regions where closure behavior remains visible across recorded observer and scale views, then return those regions to source-facing rows for inspection. The archive result in this report is a bounded example of that primitive: a repeated structural signature is observed, constrained by non-archive sensitivities, and grounded back to selected text rows without becoming a model ranking.

A concrete use case is post-hoc audit triage for a document-grounded LLM workflow. A red-team analyst or evaluation engineer could use the Gate12B surface to identify rows where a generated answer follows a conflict anchor, declines to glue available evidence, or remains structurally visible across recorded artifact views. The output would not decide correctness automatically; it would prioritize which prompt/answer/source rows deserve human inspection before the workflow is summarized as pass/fail, preference, or accuracy.

8 Threats to Validity

The main threats to validity are:

- **Existing artifacts only.** The study uses existing local artifacts rather than rerunning model inference.
- **Current dense-transformer set only.** The current evidence set contains four model lines.
- **Three rendering families only.** The family comparison covers `archive_v1`, `transcript_v1`, and `briefing_v1`.
- **Selected source queues.** The current source-facing queues use `per_band_limit=2`.
- **Derived annotation.** Source-facing annotation is exact-anchor / phrase-rule derived, not human semantic annotation.
- **Anchor dependence.** Source-facing annotation depends on the available support-anchor and conflict-anchor text in the Gate12A text-surface artifact.
- **Candidate definitions.** Candidate extraction depends on the current observer and scale definitions.
- **Motif expansion boundary.** `cycle_motif_expansion_v1` is an explicit sensitivity, not the default observer set.
- **Bounded reparameterization.** The reparameterization result is limited to the evaluated `basis_coordinate_reversal_v1` transform.

- **Packaging boundary.** Generated `runs/` artifacts are not committed into this paper source directory.
- **No causal inference.** The artifact result does not imply a weight-level causal mechanism.

The source-facing alignment is intentionally small. The selected archive queue contains 16 rows, and the strongest high-side annotation statement is 8/8 rows. This is sufficient for a bounded inspection result, but not for a statistical generalization. Larger `per_band_limit` queues and supplied human annotation are future validation work.

This report also does not claim a residual effect-size result. The primary reported surface is the relation-signature direction and its sensitivity behavior. Residual distribution summaries should be added in a later evidence package before making any effect-size claim.

These limitations do not destroy the bounded claim because the claim is deliberately artifact-bounded. The integrity checks and source-linkage checks are recorded in the evidence manifest. The non-archive sensitivity constrains a narrow degenerate annotation-rule-only explanation. The non-claims are explicit, and the paper does not ask the result to support a broader conclusion than the current artifact evidence earns.

Table 9: Non-claim boundary.

Not claimed	Why the boundary matters
Universal law	The evidence is a bounded artifact report over selected model lines, families, and runs.
Model-quality benchmark	Gate12B reads closure signatures; it does not rank models or score answer quality.
Correctness classifier	High-tension and flat are candidate sides, not correctness labels.
Weight-level claim	The report does not infer causal mechanisms inside model weights.
Physical invariant	The reparameterization check is an artifact-level stability check over the evaluated transform.
Gate12A checkpoint claim	The manuscript is a bounded technical report, not a Gate12A checkpoint or replacement.

9 Artifact Manifest and Reproducibility Boundary

The canonical evidence manifest path is `workstream/229_GATE12B_PAPER_EVIDENCE_PACKAGE_MANIFEST.md`. The manuscript records a manifest-level evidence map. Generated `runs/` artifacts are not committed into this paper source directory.

Table 10: Claim-to-artifact map.

Claim fragment	Evidence set	Integrity line
Archive high R / flat M repeats across four model lines	Motif specificity summary and 36 motif Gate12B runs	36/36 checksums present; mismatch count 0
Archive direction is observed under the representative motif path and scale-focused support path; core-observer boundary path is a candidate-vanish boundary	Motif archive runs plus archive boundary runs	24 boundary runs with checksums present; mismatch count 0
Archive source-facing alignment	Archive source queue and annotation artifacts	16 queue rows / 16 annotation rows; checksum mismatch 0
Transcript/briefing do not reproduce clean archive alignment	Non-archive source queue and annotation artifacts	transcript 16/16 rows; briefing 16/16 rows; checksum mismatch 0
Queue rows are linked to the matching text-surface source	Source queue manifests and text-surface manifests	12/12 cases have matching source_gate12a_run_id

The required evidence sets are:

- motif specificity summary
- 36 motif Gate12B runs
- archive, transcript, and briefing source queue artifacts
- archive, transcript, and briefing source-facing annotation artifacts
- archive boundary runs for appendix sensitivity

Recorded integrity status in the manifest:

- 36 motif Gate12B runs: checksum mismatches 0
- 6 source queue / annotation artifacts: checksum mismatches 0
- 24 archive boundary runs: checksum mismatches 0
- 12/12 source queue cases: matching Gate12B/text-surface **source_gate12a_run_id**

The manifest is maintained as a separate workstream file rather than embedded in the manuscript source. This paper does not commit generated **runs**/ artifacts. The manifest records the artifact directories, manifests, checksums, and queue outputs intended for a future public evidence package. The claim in this manuscript is limited to the recorded manifest-level evidence mapping above.

10 Conclusion

Gate12B reports a bounded archive-family closure-signature phenomenon in existing LLM audit artifacts. The current dense-transformer archive surfaces show a repeated high-R / flat-M relation signature. The archive-family direction is observed under the representative motif path and the scale-focused support path, while the core-observer boundary path is reported as a setting where candidates can vanish. The artifact-level reparameterization record is reported as a bounded availability-and-outcome record, not as independent support. Selected source-facing rows align with that structural read: high-side R rows are conflict-following in the selected archive queue, while flat-side M rows are support-following or non-gluing.

The source-facing layer matters because it returns structural candidates to the text surface without turning them into model-quality labels. The result is therefore not just a relation-signature table, but also not a general answer evaluator. It is a bounded bridge between a closure-signature candidate surface and selected prompt/answer/anchor rows.

The boundaries remain central. The report does not claim a universal interpretability law, a correctness classifier, a model-quality benchmark, a physical invariant, a weight-level causal mechanism, or a Gate12A checkpoint result. Transcript and briefing sensitivities do not reproduce the same clean archive alignment, which supports a family-specific reading while also keeping the archive result bounded.

Future work should assemble a public evidence package, add broader source-facing annotation coverage, and test whether the archive-family signature persists under additional model lines, rendering families, and artifact transforms. We do not introduce those extensions here; the present report closes on the bounded manifest-level evidence mapping above.

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A Compact Evidence Package Table

Table 11 summarizes the compact evidence package supporting the reported claims. Generated run artifacts are not reproduced in full in this manuscript. The workstream manifest records the artifact paths, checksum status, row counts, and source-linkage checks used by the tables above.

Table 11: Compact evidence package summary.

Evidence set	Supports	Recorded integrity status
Motif specificity summary	Family-level relation-signature comparison across archive, transcript, and briefing surfaces	Summary artifact paths and hashes recorded in <code>workstream/229_GATE12B_PAPER_EVIDENCE_PACKAGE_MANIFEST.md</code>
36 motif Gate12B runs	Motif evidence surface supporting family comparison across four model lines, three families, and three <code>top_k</code> settings	Checksums present for 36/36 runs; mismatch count 0
Archive source queue and annotation artifacts	Selected archive high-R / flat-M source-facing alignment	16 queue rows and 16 annotation rows; checksum mismatch count 0
Transcript source queue and annotation artifacts	Non-archive source-facing sensitivity for <code>transcript_v1</code>	16 queue rows and 16 annotation rows; checksum mismatch count 0
Briefing source queue and annotation artifacts	Non-archive source-facing sensitivity for <code>briefing_v1</code>	16 queue rows and 16 annotation rows; checksum mismatch count 0
Archive boundary runs	Scale-focused support path, core-observer boundary path, and motif observer recovery checks	24 boundary runs with checksums present; mismatch count 0
Source queue linkage checks	Validation that Gate12B candidate runs and Gate12A text-surface audits refer to the same source artifact	12/12 source queue cases have matching <code>source_gate12a_run_id</code>

The manifest is maintained as a separate workstream file rather than embedded in the manuscript source. This paper does not commit generated **runs/** artifacts. The manifest records the artifact directories, manifests, checksums, and queue outputs intended for a future public evidence package. The claim in this manuscript is limited to the recorded manifest-level evidence mapping above.